

**A PROPOSED OFFERING OF
A VILLAGE RESIDENT INFORMATION & PROFILING SYSTEM
FOR DISIPLINA VILLAGE, UGONG**

A Project Proposal Presented to the
Faculty of Datamex College of Saint Adeline, Inc.

In Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Information Technology

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PROJECT PROPOSAL

INTRODUCTION

The Village Resident Information & Profiling System is created to make managing resident and household records easier and more organized. Many village offices still rely on logbooks, index cards, or Excel files, which can be slow and prone to mistakes. This system replaces those outdated methods with a faster and more reliable digital process. It allows users to store, update, and access information securely with just a few clicks.

One of the main goals of this system is to centralize all resident and household information in one place. It helps ensure that every record is accurate, complete, and easy to retrieve whenever needed. By digitizing the process, it reduces human error, duplication, and data loss. This makes it easier for officials to manage daily operations more efficiently.

The system also uses a role-based access feature that separates the responsibilities of the admin and staff. The office staff assist residents who come to claim or update their information, making sure everything is verified and correct. After verification, the admin inputs and manages the data inside the system to keep all records organized. This setup improves data security and ensures that only authorized users can handle sensitive information.

Aside from storing data, the system also lets admins group individuals into households, track statistics, and generate reports. It helps monitor population growth and changes within the community. With accurate and updated information, village officials can make better decisions and respond faster to residents' needs. Overall, the system saves time, reduces paperwork, and makes community record-keeping more efficient and secure.

CLIENT INFORMATION

Name of the Client Organization: Disiplina Village Ugong Valenzuela City Government

Key Contact: Office of Barangay Chairman, Disiplina Village Ugong

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Disiplina Village Ugong is a socialized in-city relocation housing project launched by the Valenzuela City Government in 2009 in response to Typhoon Ondoy. It provides safe, affordable rental housing for informal settler families (ISFs) previously living in high-risk areas. Located on a 1.9-hectare city-donated site, it initially housed around 890 families.

The project is Valenzuela's flagship housing initiative, institutionalized by Ordinance No. 36 (Series of 2011), and follows a multi-stakeholder model with support from national agencies and private partners. Beyond housing, it promotes long-term community development, low-cost living, and values-based governance through the Home Space Agreement system.

PROJECT SCOPE

The Village Resident Information & Profiling System is designed to help Disiplina Village staff manage resident and household records in a faster and more organized way. Built with secure login, editable profiles, reporting, and analytics, it replaces manual logbooks and spreadsheets with a reliable digital platform. Using the Agile approach, the project ensures step-by-step development, continuous feedback, and improvements tailored to the community's needs.

This project aims to build an easy to use system, Village Resident Information & Profiling System made specifically for Disiplina Village in Valenzuela City. It will help barangay staff manage resident and household records in one organized and accessible place.

With this system, staff can quickly update information, track family groupings, and create useful reports like population growth. It's designed to make their daily work easier and more efficient replacing the hassle of paperwork and spreadsheets. The system also helps prevent issues like duplicate records, missing data, or lost files, giving officials more time to focus on serving the community.

SECURE LOGIN AND REGISTRATION SYSTEM

- Username
- Password
- (Admin)

EDITABLE RESIDENT PROFILES:

- (Resident ID)
- FullName
- Gender
- Age
- Date Of Birth
- Address
- Civil Status
- Voter Register
- HouseHold Role
- Head Of The Family
- Spouse
- Children
- Parents
- Relative
- Others
- Admin Dashboard with full of (Total Of Resident, Total Of Household, Total Of Voters,) access
- Simple analytics for demographics and household counts
- Responsive design and desktop use

INCLUSIONS

- Frontend and backend system – The system has a user-friendly interface for interaction (frontend) and a secure backend to handle data processing and storage.
- Resident and admin roles – It supports two types of users: residents who can register and update their information, and admins who can manage records and oversee the system.
- Data validation and user authentication – Built-in checks ensure that information entered is accurate, while login authentication keeps the system secure and accessible only to authorized users.

EXCLUSIONS

- National ID/Biometric integration – Not included, as the system will focus only on basic resident profiling without advanced identification features.
- Mobile app version – A mobile application is excluded since the project will only be available through a web-based platform.
- Payment or billing modules – Excluded because the system is designed solely for profiling residents and households, not for handling financial transactions.

ASSUMPTIONS

- Internet Access for Residents and Staff – Not required because the system is fully offline.
- Admins trained to operate the system – Administrators will be provided with training so they can manage and maintain the system effectively.

CONSTRAINTS

- Limited hardware resources – The system must run efficiently on standard, non-high-end computers.
- Potential Internet Interruptions – Not applicable, since the system is offline.
- Budget limits on maintenance and hosting – Ongoing support and server costs are restricted by financial constraints.

PROJECT APPROACH

To achieve the goals of the Village Resident Information & Profiling System, the main goal is to focus on creating a system that helps village staff with their daily tasks in an easier and more organized way. We aim to create a platform where they can easily manage resident profiles, household groupings, and unit availability all in one place. Instead of relying on logbooks or spreadsheets, this system will offer a faster, safer, and more organized way of handling community data.

METHODOLOGIES AND FRAMEWORKS

We will use the Agile approach, which means we'll work on the system step by step through short phases. In each phase we focus on building specific parts of the system, then we check how it works and ask for feedback. This way, we can improve the system step by step and make sure it matches what the users really need. Agile Approach also helps us stay organized, adjust if there are changes.

ADVANTAGE OF AGILE

The system will be developed using the Agile approach, allowing continuous feedback, faster issue resolution, and adaptability to new needs. The process includes requirement gathering, UI/UX design, database setup, module development, testing, and final deployment. This ensures a reliable and user-friendly platform for managing resident and household records in Disiplina Village.

- Faster identification and fixing of issues before they get too big.
- Easy to adjust and improve based on new needs or unexpected problems.
- Frequent feedback from village staff to improve the system step by step.
- Early release of working features, so the system can be tested and used right away.

Requirement Gathering - We researched how barangay office staff usually manage resident records and the common problems they face.

UI/UX Wireframes - We'll create a simple design, so the staff can imagine how the system will look.

Database & Backend Setup - We'll create a secure place where all the data will be stored and processed, making sure it's fast, safe, and reliable.

Resident Registration Module - We'll let the staff register and manage resident details and household groupings in just a few clicks no more logbooks or scattered Excel files.

Admin Dashboard - Create a dashboard where admins can easily access Total of resident, Total of household, Voter Status.

Testing & Feedback Iteration - Let barangay staff try out the system and give feedback, then improve the system based on their experience and suggestions.

Final Deployment - Launch the completed system for official use in the village office

PROJECT TEAM

Our team worked together closely to design and develop a system that could truly help the village staff manage resident information better. Each of us played a role in building the system from planning and designing to coding and testing.

	Factor, Carlo C. Role: UI/UX Designer/ System Analyst/Data Gatherer	Planned the system flow, managed the team schedule, and created user-friendly interfaces for easy navigation and access by barangay staff. Additionally, analyzed user requirements and ensured that the system features aligned with the needs of the barangay.
	Din Samantha Rose. Role: Developer / Programmer	Built the system using Microsoft Visual Basic.Net 2010(VB.NET) and integrated the database SQL Server Management Studio. Ensured that the system worked well on barangay level hardware and ran offline with backup options.

	Sto Domingo, Joshua Role: System analysis /Data Gatherer	Met with key stakeholders to gather insights and requirements. Designed data flow diagrams and user processes to ensure the system matched real world needs.
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PROJECT TIMELINE

The Software Development Life Cycle (SDLC) as our guide throughout the project. The timeline helped us stay organized and ensured that each step was completed properly before moving to the next.

Phase	Title	Start Date	End Date	Key Activities/ Deliverables
Phase 1	Planning	July 10, 2025	July 16, 2025	We decided on what system we will make
Phase 2	Requirement Gathering	July 18, 2025	July 25, 2025	We started to talk or find information about our system
Phase 3	Design	July 27, 2025	August 9, 2025	We create a layout for the system
Phase 4	Development	August 9, 2025	August 30, 2025	We started building the actual system
Phase 5	Testing and Review	August 30, 2025	September 6, 2025	We checked if the system is working
Phase 6	Deployment	September 6, 2025	September 13, 2025	We launched the system

PROJECT RESOURCES

This project was built by a small but passionate team of students who genuinely wanted to help improve how the barangay manages resident information. We worked side by side brainstorming, designing, coding, and testing every part of the system to make sure it fits the actual needs of Disiplina Village.

We each brought different skills to the table. By combining our strengths, listening to the stories of barangay staff, and observing how they currently work, we were able to create a system that's not only functional, but also practical and easy to use in real life.

Even with limited resources, we made sure the system runs smoothly offline, works on common barangay computers, and keeps resident data safe and organized. More than just a school requirement, this became a meaningful project that we hope can make daily tasks in the barangay a little easier

HARDWARE

These hardware specifications are recommended because they are affordable, reliable, and sufficient for running Visual Basic .NET 2010 and . A Core i5 with 8GB RAM and SSD ensures smooth performance for storing and managing resident and household records, while Windows 10 provides a secure and stable platform for daily use in the barangay

- Processor: Intel Core i5 (3rd–4th Gen) or AMD Ryzen 3
- Memory (RAM): 8 GB
- Storage: 256 GB SSD (for faster boot and data access)
- Graphics: Integrated Intel HD 4000 or higher
- Operating System (OS): Windows 10 Pro (64-bit)

SOFTWARE

Microsoft Visual Basic 2010 (VB.NET) - This will be used to develop the system where all the functionalities like resident profiling, report generation, and household tracking will be built.

DATABASE MANAGEMENT SYSTEM (DBMS)

SSMS – This will store all the important data such as resident information, household groupings, login accounts, and community statistics in an organized and secure way.

OPERATING SYSTEM

Windows 10 or Higher – A stable operating system that will support both the development process and the actual use of the system on the barangay's computer

HUMAN RESOURCES

The project team is composed of members with specific roles to ensure the system's success. The Project Leader manages progress, the UI/UX Designer ensures usability, the Developer codes the system, the System Analyst aligns it with user needs, and the Data Gatherer provides accurate resident information.

Project Leader – Making sure everything stays on track and everyone knows what to do.

UI/UX Designer – Design how the system looks, making sure it's easy for the admin staff to use and navigate.

Developer/Programmer – Responsible for writing the system's code and ensuring all features function properly.

System Analyst – Understands the needs of the users and ensures the system meets them.

Data Gatherer – Collects accurate resident and household data for the system.

RISK MANAGEMENT

The project governance framework for developing the Village Resident Information & Profiling System was established to make the project move smoothly, remain focused on its objectives, transparency, and accountability. The framework concisely delineated the roles, top - down process, beginning with the Barangay Council in coordination with the project development team. Regular meetings will be conducted to review progress, address issues, and approve changes or improvements.

The design basis and core system features were rooted in the original requirements obtained from Village Resident Information & Profiling System operations and staff. The project's decision-making is led by the Barangay Council, which approves key aspects like system rollout and budget. A Project Manager supervises daily operations to keep development on track. IT Developers and Analysts handle the technical side, including system design and testing. Lastly, Barangay Staff participate in testing and training to ensure the system fits real world needs, with their feedback helping guide improvements.

Limited Technical Knowledge Risk: Barangay staff may not be comfortable using computers or new software.

Plan: We designed a very simple and English friendly interface with labeled buttons and straightforward steps.

Data Loss or System Errors Risk: Since the system is offline and stores everything locally, files could be deleted or the computer might crash.

Plan: We added a manual backup feature. Staff were trained to regularly save backups to USB drives or separate folders, making it easier to recover data if something goes wrong.

Old or Slow Barangay Computers Risk. The barangay might be using older hardware that can't handle modern software.

Plan: The system was built in VB.NET and optimized to run smoothly even on computers with as little as 8 GB RAM and Windows 10.

Unauthorized Access Risk: If anyone can use the barangay PC, they might access or edit sensitive resident data.

Plan: The system uses login credentials with an admin role, ensuring that only authorized administrators can access and update resident records.

Software Bugs or Unexpected Errors Risk: Like any program, bugs or issues might show up during actual use.

Plan: We tested the system using sample resident data from the barangay and fixed any problems we found. The system is also built to be easily updated if needed in the future

COMMUNICATION PLAN

Effective communication will be essential to the success of the Village Resident Information & Profiling System. Because the system was being developed by our project team with representatives from Barangay Council operations and staff. Communication throughout the project will be managed using structured updates, regular coordination meetings, and open channels between stakeholders. The goal is to ensure that all parties are aligned, informed, and able to respond to developments or challenges quickly.

Monthly Stakeholder Meetings (In-Person):

- Key stakeholders including the Barangay Captain, Project Manager, Lead Developer, and Barangay Secretary meet once a month to review progress, discuss achievements, approve changes, and address major issues.
- Emergency or Urgent Meetings (As Needed): When critical issues arise, such as data breaches, system downtime, or urgent concerns, emergency meetings may be held either virtually or onsite depending on availability.

KEY STAKEHOLDERS AND THEIR COMMUNICATION PREFERENCES

The project's communication and governance approach makes sure everyone stays connected and involved in a way that works best for them. From the Barangay Council down to the staff and residents, each group has clear roles and gets updates through the channels they're most comfortable with. This helps the system run smoothly, keeps things transparent, and makes sure the new system truly supports the community's needs.

Communication Tools

- Messenger – Used for quick chats and easy sharing of updates within the team.
- Viber – Provides an alternative messaging app for group communication and file sharing.
- Google Meet – Used for online meetings and discussions when face-to-face interaction isn't possible.
- Emergency or Urgent Meetings - A way to call and record immediate meetings so important issues can be discussed and acted on without delay.

Type of Communication	Frequency	Format/Tools Used	Purpose
Messenger	Daily	Chat messages	For quick updates, reminders, and sharing small files among the team
Viber	2–3 times a week	Group chat / file sharing	For group discussions and sharing project-related documents
Google Meet	Weekly or as needed	Video meetings	For virtual meetings to discuss progress, assign tasks, and resolve issues
Emergency or Urgent Meetings	As Needed	Virtual (e.g., Messenger, Viber, Googlemeet Teams) or Onsite	Address critical issues such as data breaches, system

Type of Communication	Frequency	Format/Tools Used	Purpose
			downtime, or urgent concerns

PROJECT GOVERNANCE

The project governance framework for the Village Resident Information & Profiling System ensures smooth progress, focus on objectives, transparency, and accountability. It outlines a top-down process led by the Barangay Council in coordination with the project team. The Council makes key decisions on rollout and budget, while the Project Manager oversees daily operations. IT Developers and Analysts manage system design and testing, and Barangay Staff are involved in training and feedback to align the system with real-world needs. Regular meetings support progress reviews, issue resolution, and improvements

Roles and responsibilities

Role	Responsibilities
Client Representative	Provides project requirements, approves major changes, and receives progress updates.
Project Leader	Oversees the project, coordinates stakeholders, and ensures deadlines and quality standards are met.
System Analyst	Analyzes requirements, designs system specifications, and ensures alignment with business goals.
Data Gatherer	Collects accurate data, conducts surveys or interviews, and ensures data integrity for reporting.
Software Engineer	Writes and tests code, implements features based on specifications, and reports progress to the Project Leader.
UI/UX Designer	Designs the user interface, creates layouts, and ensures the system is user-friendly.
Tester	Conducts system testing, identifies bugs, and ensures functionality and performance meet requirements.

APPENDIX

- **Appendix A: Related Research / Studies (RRS)**

Author / Year	Study Title	Findings / Features	Relevance to Study
Turaeva, N., Mukhiddinov, B., & Nematov, Z. (2021)	Development of electronic system for registration of citizens living in multi-storey buildings.	Proposed an electronic registration system for residents in multi-storey buildings, focusing on efficient record-keeping and accessibility.	Provides a model for digitizing resident data that can be adapted for barangay profiling systems.
	International Journal of Advanced Trends in Computer Science and Engineering, 10(4), 2523–2527. Link		
Coursera Staff (2025, July 22)	<i>What Is Agile? An iterative approach to project management.</i>	Explains Agile methodology as an iterative project management and development approach.	Offers guidance on adopting Agile methods for system development, ensuring flexibility and adaptability.
	Coursera. Link		
Lacasandile, A. D., Abisado, M.	Designed a barangay information	Serves as a strong reference for creating a resident profiling system with a	

Author / Year	Study Title	Findings / Features	Relevance to Study
B., Labanan, R. M., & Abad, L. P. (2020, August)	dashboard that supports decision- making and improves local governance through digitized records.	decision-support dashboard.	
Housing and Urban Development Coordinating Council (HUDCC). (n.d.)	<i>Disiplina Village, Bignay, Valenzuela City. Link</i>	Presents a case of organized housing and community management in an urban resettlement area.	Demonstrates real-world applications of structured resident management relevant to barangay profiling.

- **Appendix B: Group Members' Task Distribution**

Name	Role(s)	Assigned Task(s)
Factor, Carlo C.	UI/UX Designer, Project Leader, System Analyst	Designs the user interface, leads the project, and analyzes system requirements.
Sto. Domingo, Joshua.	System Analyst, Data Gatherer	Analyzes requirements and collects accurate project data.
Din, Samantha Rose.	Software Engineer	Writes and tests code, implements features, and supports system development.